



**Mansoura University**  
**Faculty of Computers and Information**  
**Fall 2023-2024**



# **DECISION SUPPORT SYSTEMS**

**Grade: 4<sup>TH</sup> YEAR (GENERAL)**

**Prepared by:**

**Ibrahim El-Hasnony**

## QUIZ-I

- 1) .....is an umbrella term that combines architectures, tools, databases, analytical tools, applications, and methodologies (DSS, BI, Data mining, MSS).
- 2) The process of ..... is based on the transformation of data to information, then to decisions, and finally to actions. (DSS, BI, Data mining, MSS).
- 3) Managers and ..... specialists use the concept of a system frequently and yet it is hard for most of us to define and understand the concept.(EIS, TPS, MIS, DSS).
- 4) Systems also have ..... mechanisms to provide a means of controlling the operation of the system.( input, output, feedback, results).
- 5) Feedback comes from the .....( DSS, Decision maker, Customer, System).
- 6) In a Decision Support System, the primary focus is often on the computerized components of the system ( T/F)

## QUIZ-2

- 1) Information systems capture data from the organization (internal data) and its environment (external data). (T/F).
- 2) Depending on the type of ....., the information output may take the form a query response, decision outcome, expert-system advice, transaction document, or a report. (DSS, MIS, IS, ES).
- 3) Most of the data captured by information systems relates to the operations of the organization itself. (T/F).
- 4) The objectives of a system are realized in its .....(input, output, feedback, testing).
- 5) The fewer resources a system consumes in producing given outputs, the more effective it is.(T/F).
- 6) The objective of management reporting systems is to provide routine information to managers (T/F).

## QUIZ-3

- 1) .....support top managers with conveniently displayed summarized information, customized for them. (MIS,DSS,TPS,EIS)
- 2) .....measuring performance against the planned objectives and initiating corrective action, if needed.(planning, leadership, controlling, organizing)
- 3) .....clarifies the computer's role in aiding rather than replacing the decisionmaker. (Decision, support, systems, DSS).
- 4) The decision support system is under user control, from early beginning to final putting into use and daily use (T/F).
- 5) DSS refers to a class of systems, which supports the process of decision-making and does not always give a decision itself (T/F).
- 6) .....consists of various mathematical and analytical models that are used to analyze complex data, thereby producing the required information. (DSS DB, DSS SW, DSS KMS, DSS Dialogue)
- 7) To increase the production level by 40 percent using the backward sensitivity analysis, first, the target value for the production level can be set. (T/F)

## QUIZ-4

- 1) Its purpose is to help conduct a meeting, or for users to collaborate ( office, data , communication, document) driven DSS.
- 2) An Inter-Organizational DSS provides stakeholders with access to a company's intranet and authority or privileges to use specific DSS capabilities. (T/F).
- 3) ERP is NOT a Decision Support System (T/F).
- 4) ..... Decisions are those in which the decision maker must provide judgment, evaluation, and insights into the problem definition (Structured, semi-structured, unstructured).
- 5) .....involves identifying and exploring various solutions to the problem.(intelligence, design, choice, implement).
- 6) The set of possible alternatives is the output of the .....phase. (intelligence, design, choice, implement).

## QUIZ-5

- 1) Every time you park downtown, you get a parking ticket because you exceed the time limit on the parking meter although once it did not happen, this is an example for decision making under.....(**certainty, risk, uncertainty**).
- 2) In the payoff table, the rows correspond to the possible decision alternatives.(**T/F**).
- 3) .....is the increase in the expected profit that would result if one knew with certainty which state of nature would occur.(**EV, EVPI, Regret, Optimistic**).
- 4) In pro forma financial model Costs are estimated based on past data, gross sales are predicted, and profit or loss is then calculated on these relationships (**T/F**).
- 5) .....models alters the historical data mathematically to better reflect the assumptions of a decision maker.(**exponential smoothing, time series, regression, moving average**)
- 6) DSS with simulation models conduct experiments to identify conditions or situations that approximate the actual conditions (**T/F**).

## QUIZ-6

- 1) .....is an approach to discovery and problem solving by employing practical methods (**Heuristics, inference engine, Knowledge acquisition, case reasoning**).
- 2) .....used to determine the distance between or relationship among various components. (**Heuristics, inference engine, Knowledge acquisition, case reasoning**).
- 3) “What techniques will it use?” used in .....for designing web-based DSS.(**conceptualization, problem identification, feasibility analysis, SW development**).
- 4) The passive data dictionary (also called integrated data dictionary) is the one used only for documentation purposes. (**T/F**).
- 5) .....is an organized approach to design specialized decision support systems, more precisely their user interfaces.(**ROMC, ACID, EVPI, RANI**).
- 6) .....means that a decision support system must be resourceful enough to perform the entire range of tasks that a decision maker needs to perform when making a decision using DSS. (**Adaptability, Versatility, Quality, Fatigue**).

# COURSE DETAILS

## Course Syllabus

**Chapter 1:** Introduction to DSS

**Chapter 2:** Decision Making Concepts

**Chapter 3:** Decision Making Technologies and Environment

**Chapter 4:** Model Management (MDSS)

**Chapter 5:** Data Management System and Dialogue Management

**Chapter 6:** Designing and Developing Decision Support Systems

**Chapter 7:** Executive Support Systems

**Chapter 8:** Automated Decision Systems and Expert Systems

**Chapter 9:** Knowledge Management and Collaborative Systems

**Chapter 10:** DSS Intelligent Systems & Tools and Techniques

**Chapter 11:** Business Intelligence: Data Warehousing, Data Acquisition, Data Mining, Business Analytics, and Visualization



## CHAPTER 6 CONTENTS

- Designing Issues
- Decision oriented analysis
- DSS audit plan

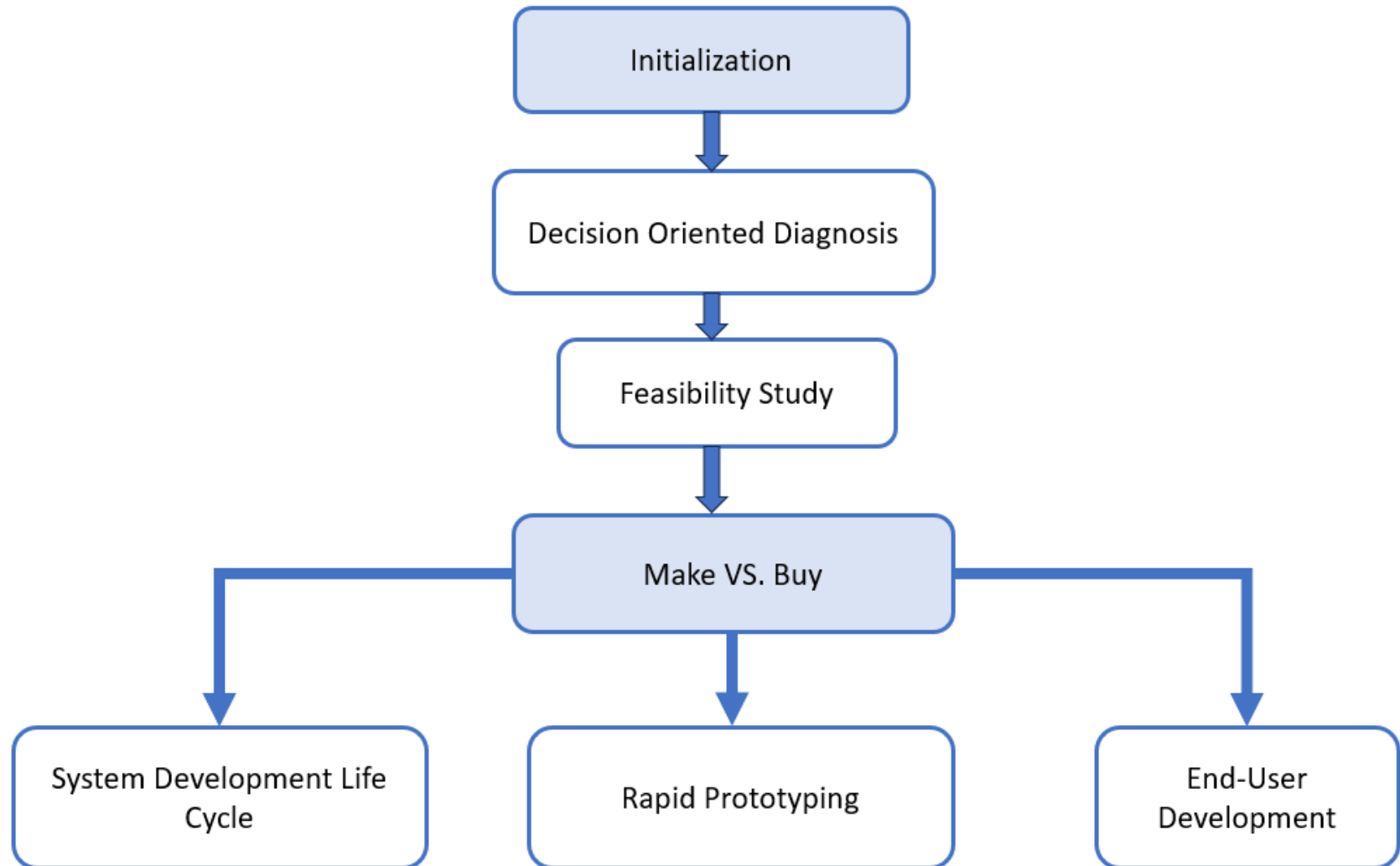
# DESIGNING ISSUES

- **Designing and development of a decision support system is a complex process, you may face several issues:**
  - 1) Mismatch between the perspectives of the programmer and decision maker (communication, priorities, risk).
  - 2) Identification of specific requirements
  - 3) Technology selection
  - 4) Approach to software design and development

## ISSUES POSSIBLE SOLUTIONS

- **Most of these issues can be resolved by:**
  - 1) Identifying clear agendas
  - 2) Brainstorming with team members
  - 3) Communicating the expectations clearly to the programmers
  - 4) Educating yourself about various technologies used in DSS development.
  - 5) Trying to reach a common platform with the developers.

# DSS DESIGN PROCESS



# INITIALIZATION

- The initialization phase involves **defining the purpose and objectives** of the DSS.
- This is where you identify the specific decision-making **problems** or **opportunities** that the DSS will address.

## DECISION-ORIENTED DIAGNOSIS (I)

- During the diagnosis phase, you conduct a **thorough analysis** of the **current** decision-making processes within the organization.
- This includes understanding **how decisions are made**, the data and information **required**, and the **pain points** or **inefficiencies** in the existing system.
- This step often involves conducting **interviews**, **surveys**, and **data analysis** to pinpoint areas where the DSS can make a significant impact.

## DECISION-ORIENTED DIAGNOSIS (2)

- **Diagnosis** is the identification of problems or opportunities for improvement in current decision behavior.
- Diagnosis involves determining **how** decisions are currently made, specifying **how** decisions should be made, and understanding **why** decisions are not made as they should be.
- These statements of improvements provide the objectives for the DSS development.

## DECISION-ORIENTED DIAGNOSIS (3)

- **Diagnosis of a decision process involves completing the following three activities:**

- 1) Collecting data on current decision-making using techniques such as interviews, observations, questionnaires and historical records;
- 2) Establishing a coherent description of the current decision process.
- 3) Specifying a norm for how decisions should be made.



## DECISION-ORIENTED DIAGNOSIS (4)

- A related diagnostic activity is conducting a **DSS Audit**.
- In general, it can be very useful to **audit operational and managerial decision processes**.

- ✓ In the context of a "DSS audit plan," the term "audit" refers to a systematic and comprehensive examination or review of the Decision Support System (DSS) and its various components to assess its performance, effectiveness, security, compliance with regulations, and overall health.
- ✓ The primary purpose of an audit is to identify strengths and weaknesses in the DSS and to provide recommendations for improvements.

## DSS AUDIT PLAN (I)

- **A Decision Support System (DSS) audit plan** is a structured approach for evaluating and assessing the effectiveness, efficiency, and security of a Decision Support System within an organization.
- DSS audit plans are essential for ensuring that the DSS is functioning as intended, providing accurate and reliable information, and complying with relevant regulations and standards.

## DSS AUDIT PLAN (2)

|        | DSS Audit Plan                                                                                                                                                                                                                                                             |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Step1  | Define the decisions, decision processes and related business processes that will be audited. Define the authority of the auditor, purpose of the audit, scope of the audit, timing of the audit, and resources required to perform the audit. Identify a primary contact. |
| Step 2 | Examine the formal design of the process. Diagram the process and specify criteria, etc. Is the design effective and efficient?                                                                                                                                            |
| Step 3 | Examine the actual use of the decision process. Observe the process. Interview decision makers and collect data. Is the process implemented and used as intended?                                                                                                          |
| Step 4 | Assess performance of the actual decision process. What works? Can cycle time be reduced? Are decisions appropriate? Timely? Cost effective? Is the process producing value in meeting business objectives? If not, why?                                                   |
| Step 5 | Reporting and recommendations. Summarize steps 1-4 in a written report. Discuss what is working well and what needs to be improved. Develop recommendations for improving the process. Hold an exit meeting with decision makers.                                          |

## PREPARE A FEASIBILITY STUDY (I)

- This study should occur prior to committing resources to developing a proposed DSS.
- Involves assessing the technical, economic, operational, and scheduling feasibility of the project.
- You need to determine if the organization has the resources, budget, and technical capabilities to develop and maintain the DSS. A cost-benefit analysis is often conducted to evaluate the potential return on investment.

## PREPARE A FEASIBILITY STUDY (2)

- What should be included in a DSS Feasibility Study? like DSS Scope and Target Users, Anticipated DSS Impacts, Benefits, Risks and Mitigating Factors.
- At this point a decision is made between purchasing an application package and in- house development.

## MAKE OR BUY

- In this step, you decide whether to build the DSS in-house or purchase an existing DSS solution from a vendor.
- This decision **depends** on factors like the organization's **expertise**, **budget**, and **time** constraints.
- Buying a DSS may save time and resources, while building one in-house allows for more customization and control.
- Packaged DSS applications are often quite versatile and are usually less expensive to implement than in-house development.
- Packaged solutions are also often faster to implement.

# CHOOSE A DEVELOPMENT APPROACH

- **There are three main approaches to system design and development:**
  - 1) The traditional system development life cycle (SDLC) approach;
  - 2) Rapid prototyping and
  - 3) End-user approach.
- You may want to choose a methodology that ensures that the final product accomplishes your objectives

# SYSTEMS DEVELOPMENT LIFE CYCLE APPROACH

■ The systems development life cycle (**SDLC**) approach is based on a series of formal steps, including the following seven steps:

- 1) Confirm user requirements.
- 2) Systems analysis.
- 3) System design.
- 4) Programming.
- 5) Testing.
- 6) Implementation.
- 7) Use and Evaluation.



## RAPID PROTOTYPING (I)

- Creating a rapid prototype for a Decision Support System (DSS) design involves **quickly building a simplified version** of the system to demonstrate its key features and functionality.
- Creating a rapid prototype for a DSS allows you to visualize and test the system's design early in the development process, helping to ensure that the final DSS will meet user needs and expectations.
- It can save time and resources by identifying issues and improvements before extensive development efforts are undertaken.

## RAPID PROTOTYPING (2)

- **A typical prototyping methodology usually includes five steps:**
  1. Identify user requirements.
  2. Develop a first iteration DSS prototype.
  3. Evolve and modify the next iteration DSS prototype.
  4. Test DSS and return to step 3 if needed.
  5. Full-scale implementation.
- Rapid Application Development (RAD) specifies incremental development with constant feedback from potential users.

## END-USER DSS DEVELOPMENT (I)

- The end-user approach emphasizes **active involvement of end-users** in the system development process.
- It's often used in **situations where the end-users have a deep understanding of their needs and can contribute directly to system design.**
- This approach can help ensure that the final system **closely aligns with user expectations.**
- End-users may be involved in requirements gathering, testing, and even some aspects of system design.

## END-USER DSS DEVELOPMENT (2)

- The major advantage of encouraging end-user DSS development is that the person who wants computer support will be involved in creating it.
- The manager/builder controls the situation and the solution that is developed.
- End-user DSS development can also sometimes result in faster development and cost savings.

## END-USER DSS DEVELOPMENT (4)

- End-user DSS development is a **controversial topic**. Information systems staffs have many concerns including:
  - 1) End-users may select an inappropriate software product as a development environment.
  - 2) The end-user may have limited expertise in the use of the product and the IT group may have limited resources to support end-user development.
  - 3) Errors during end-user DSS development are frequent.
  - 4) Unnecessary databases are sometimes developed by end-users for their DSS.
  - 5) A major quality issue involves testing and limited documentation. End-users often perform only limited testing of DSS they develop; and they have limited experience- documenting applications.
  - 6) End-user databases may be poorly constructed and difficult to maintain.
  - 7) End-users rarely follow a systematic development process.

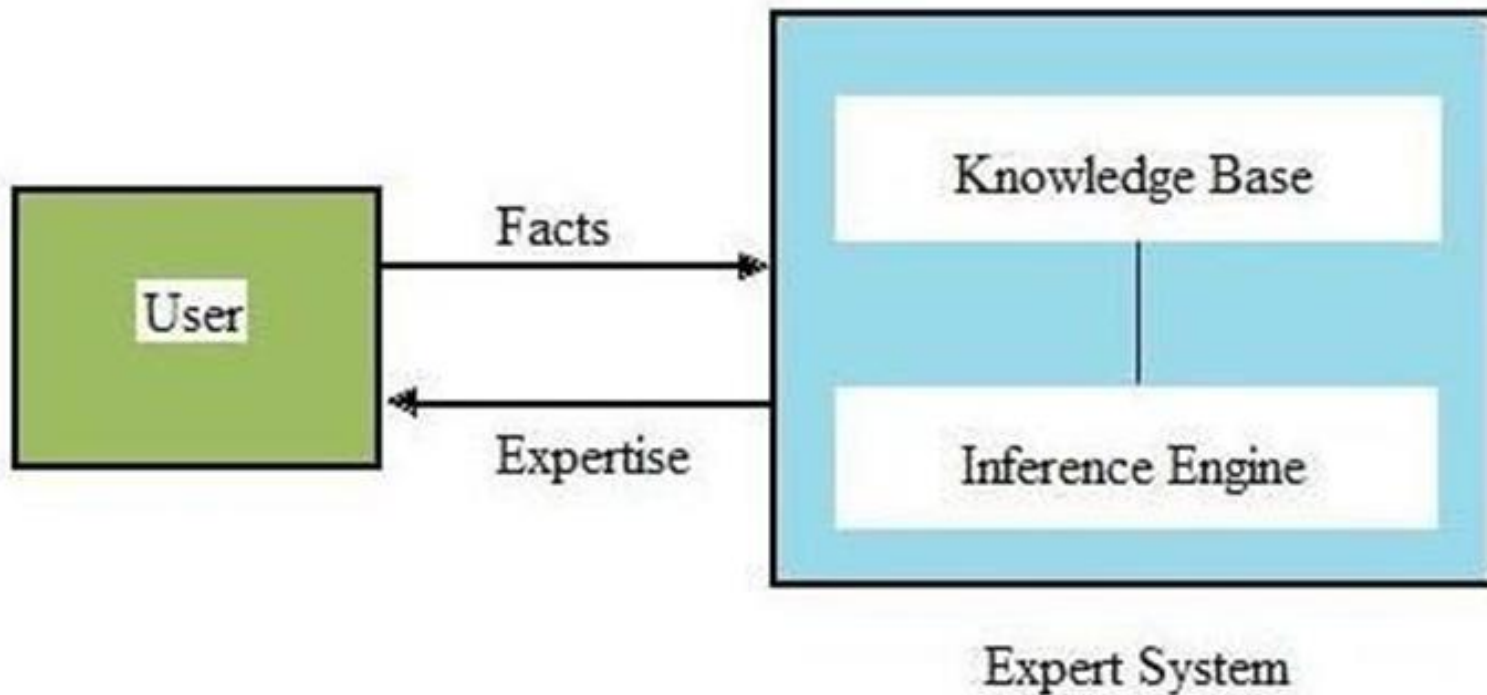
## CHAPTER 7 CONTENTS

- Components of executive support systems
- Role of ESS in firms
- Business value of ESS
- ESS for competitive intelligence

# ESS

- The ESS help managers with unstructured and semi structured problems by focusing on the information needs of **senior management**.
- Combining data from internal and external sources, **ESS create a generalized computing and communications environment that can be focused and applied to a changing array of problems.**
- ESS help senior executives monitor organizational performance, track activities of competitors, spot problems, identify opportunities, and forecast trends.
- Designed for **top-level executives and senior managers.**
- Its primary purpose is to provide **high-level, summarized information to aid in strategic decision-making.**

## ESS COMPONENTS (I)





## ESS COMPONENTS (2)

- **The knowledge base:** This is where the information is stored in the expert system in form of facts and rules. Basically, a series of IF statements. Here the programmer writes the code for the expert system.
- **The user interface:** This is where the user interacts with the expert system. This is where questions are asked and advised is produced. The user interface can also output justification features of the expert system.
- **The inference engine:** This applies the facts to the rules and determines the questions to be asked of the user in the user interface and which order to ask them. This is an invisible part of an ESS when the program is run.

# THE ROLE OF EXECUTIVE SUPPORT SYSTEMS IN THE FIRM (I)

- Bring together data from all parts of the firm and enable managers to select, access, and tailor them as needed using easy-to-use desktop analytical tools and online data displays
- ESS can drill down, moving from a piece of summary data to lower and lower levels of detail.
- The ability to drill down is useful not only to senior executives but also to employees at lower levels of the firm who need to analyze data
- Many managers have access to news services, financial market databases, economic information, and whatever other public data they may require.

## THE ROLE OF EXECUTIVE SUPPORT SYSTEMS IN THE FIRM (2)

- Include tools for modeling and analysis.
- With only a minimum of experience, most managers find they can use these tools to create graphic comparisons of data by time, region, product, price range, and so on

# BUSINESS VALUE OF EXECUTIVE SUPPORT SYSTEMS

- Much of the value of ESS is found in their **flexibility**
- Ability to **analyze**, **compare**, and **highlight** trends
- Executives are using ESS to **monitor performance** more successfully in their own areas of responsibility
- Improve **management performance** and increase upper management's span of control

# ESS FOR COMPETITIVE INTELLIGENCE

- ESS can help managers to :
  - Identify changing market conditions,
  - Formulate responses,
  - Track implementation efforts,
  - Learn from feedback.

## IDENTIFY CHANGING MARKET CONDITIONS,

- ESS can provide managers with [access to a wealth of data and information](#).
- By analyzing this data, ESS can help managers identify changing market conditions, trends, and emerging opportunities or threats.
- This information can be critical for making informed decisions and adjusting strategies in response to market dynamics.

## FORMULATE RESPONSES

- Once managers are aware of changing market conditions, ESS can assist in formulating appropriate responses.
- ESS can provide tools for scenario analysis, modeling, and decision support, helping managers evaluate different courses of action and select the most effective strategies to address the identified challenges or opportunities.

## TRACK IMPLEMENTATION EFFORTS

- ESS often includes project management and tracking features.
- Managers can use ESS to monitor the progress of various initiatives, ensuring that strategies and plans are implemented as intended.
- This tracking capability is essential for staying on course and ensuring that organizational resources are allocated effectively.



# LEARN FROM FEEDBACK

- ESS can capture and analyze feedback and performance data, enabling managers to learn from the outcomes of their decisions and actions.
- This learning process is vital for continuous improvement and adaptability.
- By using ESS, managers can gain insights into what works and what doesn't, allowing them to refine their strategies over time.



**Thanks**  
**Any questions**